VSR-PathRAG: A Verification–Scoring–Repair Framework for Trustworthy Graph-Based Retrieval-Augmented Generation

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Abstract

Retrieval-Augmented Generation (RAG) systems that ground large language model (LLM) outputs in external knowledge suffer from two fundamental trust deficits: (1) *evidence insufficiency*—the retrieved evidence may not logically entail the generated answer, and (2) *claim hallucination*—unsupported or contradicted claims can survive into the final output. Existing GraphRAG approaches retrieve relevant subgraphs but lack a unified runtime contract for deciding whether the evidence is sufficient for the answer's obligations and claims. We propose **VSR-PathRAG**, an evidence-contract-guided Verification–Scoring–Repair framework. The contract makes question obligations, claim support, evidence necessity, and submission risk explicit; MPS, CPN, and RCS are its selection, intervention, and decision primitives rather than three unrelated contributions. On UltraDomain Mix130, strict ablation over seven configurations shows that the rescue-enhanced VSR layer improves F1 from 0.2206 without repair to 0.2878 while reducing abstention. Under the frozen SAGE v33-preservation instantiation, locked unseen75 plus Repeat1–8 provide 326 zero-overlap holdout examples with complete lexical, same-rubric judge, and pairwise evaluation: SAGE v33 reaches weighted F1 0.3164 versus 0.2972 for both Naive RAG and HyDE, judge overall 3.555 versus 2.960/2.942, and average pairwise target win rates of 59.66% versus HyDE and 61.50% versus Naive. A focused mechanism audit further shows that the current contract-gap score identifies harder rows but is not yet an estimate of escalation benefit, making selective upgrade prediction the central next step.

Index Terms

Retrieval-Augmented Generation, Knowledge Graph, Evidence Verification, Counterfactual Reasoning, Risk Calibration, PAC-Bayesian Bounds

I. Introduction

A. Motivation

Large Language Models (LLMs) generate fluent but potentially ungrounded text. Retrieval-Augmented Generation (RAG) mitigates hallucination by conditioning generation on externally retrieved evidence. However, three critical gaps remain:

(1) No proof structure. Standard RAG retrieves top-k passages and concatenates them into the prompt. There is no formal notion of whether the retrieved set *proves* the answer's claims. This is analogous to presenting evidence without making a legal case.

(2) No necessity analysis. Even when evidence supports the answer, some evidence paths may be *replaceable*—removing them does not affect the answer's correctness. Without identifying *necessary* evidence, the system cannot detect fragile proofs or prioritize retrieval budget.

(3) No risk gating. RAG systems submit answers regardless of evidence quality. When the evidence is insufficient or contradictory, the system should *abstain* or *repair* rather than fabricate.

B. Contributions

We propose VSR-PathRAG, which addresses these gaps through one evidence-contract problem:

1) **Evidence Contract**: We represent a question as obligations and an answer as atomic claims, then compute a contract state describing coverage, support, contradiction, and answer-shape deficits.

2) **Deficit-Guided Escalation**: MPS and CPN provide the evidence-selection and necessity primitives used to decide whether verification or rescue is worth the additional cost.
3) **Controlled Repair and Submission**: RCS provides the final decision layer, while deterministic preservation guards prevent a repair from being accepted when it loses supported content without compensating evidence.
4) **Mechanism-Focused Evaluation**: We report a complete UltraDomain Mix130 ablation, external retrieval-baseline comparisons, representative graph-RAG supplements, locked zero-overlap holdouts with 326/326 complete rows, and an explicit audit of whether contract deficit predicts escalation benefit.

II. Problem Formulation

A. Notation

Let q denote a user question, $\mathcal{G} = (V, E)$ a knowledge graph, $\mathcal{E} = \{e_1, \ldots, e_N\}$ the retrieved evidence set, $\mathcal{O} = \{o_1, \ldots, o_P\}$ the question obligations (required answer facets), $\mathcal{C} = \{c_1, \ldots, c_M\}$ the atomic claims extracted from the generated answer, and $\Phi : \mathcal{E} \times \mathcal{C} \to \{+1, -1, 0\}$ the signed claim-evidence graph.

B. Formal Definitions

Definition 1 (Answer Trustworthiness)**.** *An answer a with claims $\mathcal{C}$ is trustworthy with respect to evidence $\mathcal{E}$ if:*

$$\forall c_i \in \mathcal{C} : \exists e_j \in \mathcal{E} \text{ s.t. } \Phi(e_j, c_i) = +1 \text{ and } \nexists e_k \in \mathcal{E} \text{ s.t. } \Phi(e_k, c_i) = -1. \tag{1}$$

Definition 2 (Proof Subgraph)**.** *A subset $S \subseteq \mathcal{E}$ is a proof subgraph for answer a if:*

$$\mathrm{cov}_{\mathcal{O}}(S) \geq \tau_O \quad \text{and} \quad \mathrm{cov}_{\mathcal{C}}(S) \geq \tau_C, \tag{2}$$

where τ_O, τ_C are coverage thresholds. S is minimal if no proper subset $S' \subsetneq S$ satisfies both constraints.

Definition 3 (Counterfactual Necessity)**.** *Evidence $e_j \in S$ is necessary if removing it causes either $\mathrm{cov}_{\mathcal{C}}(S \setminus \{e_j\}) < \tau_C$ or $\mathrm{cov}_{\mathcal{O}}(S \setminus \{e_j\}) < \tau_O$.*

Definition 4 (Calibrated Risk)**.** *The calibrated risk of submitting answer a is:*

$$R(a, S) = \sum_{i=1}^{5} w_i \cdot r_i, \quad \sum_{i=1}^{5} w_i = 1, \tag{3}$$

where each $r_i \in [0, 1]$ is a normalized risk signal: audit prior, unsupported claim ratio, proof gap, necessity gap, and contradiction risk.

Definition 5 (Evidence Contract Deficit)**.** *For question q, answer a, and evidence set $\mathcal{E}$, the evidence-contract deficit is*

$$D(q, a, \mathcal{E}) = w_O(1 - \mathrm{cov}_{\mathcal{O}}) + w_C(1 - \mathrm{cov}_{\mathcal{C}}) + w_T(1 - \mathrm{cov}_{\mathit{terms}}) + w_S g_{\mathit{shape}}, \tag{4}$$

where the weights are non-negative and sum to one, and g_{shape} measures answer-shape violations such as missing required facets or insufficient distinct points. The deficit is a runtime state used to trigger verification or rescue; it is not a gold-label correctness score.

Definition 6 (Upgrade Utility)**.** *The utility of escalation is the quality improvement obtained after spending additional retrieval or verification cost:*

$$U_{\mathit{up}} = \Delta Q - \lambda \Delta\,\mathrm{cost}, \tag{5}$$

where ΔQ is measured only for offline analysis and is not available to the runtime policy. A central empirical question is whether a reference-free estimate of D can predict positive U_{up} rather than merely predict that a row is difficult.

Problem Statement. Given q, $\mathcal{G}$, and a generated answer a, find: (1) a minimal proof subgraph S^* that maximizes joint coverage; (2) the set of necessary evidence paths $N \subseteq S^*$; and (3) a submission decision $d^* \in \{\text{submit}, \text{monitor}, \text{block}, \text{abstain}\}$ that minimizes expected untrustworthiness.

Algorithm 1 Minimal Proof Subgraph Optimization

Require: Evidence set $\mathcal{E}$, obligations $\mathcal{O}$, claims $\mathcal{C}$, budget K, B
Ensure: Minimal proof subgraph S^*

$S \leftarrow \emptyset$
while $|S| < K$ **and** $\text{tokens}(S) < B$ **do**
  $e^* \leftarrow \arg\max_{e \in \mathcal{E} \setminus S} \frac{\Delta \text{cov}_{\mathcal{O}}(e|S) + \lambda \cdot \Delta \text{cov}_{\mathcal{C}}(e|S)}{\text{tokens}(e)}$
  if $\Delta f(e^*|S) = 0$ **then**
    break {No marginal gain}
  end if
  $S \leftarrow S \cup \{e^*\}$
end while
// Minimality certificate: leave-one-out redundancy check
for each $e_j \in S$ **do**
  if $\text{cov}_{\mathcal{O}}(S \setminus \{e_j\}) \geq \tau_O$ **and** $\text{cov}_{\mathcal{C}}(S \setminus \{e_j\}) \geq \tau_C$ **then**
    $S \leftarrow S \setminus \{e_j\}$ {Prune redundant}
  end if
end for
return S

III. Evidence Contract Construction

The evidence contract is instantiated by making the question obligations and answer claims explicit, then selecting a compact evidence set that covers both. MPS is therefore not an isolated graph algorithm in the system; it is the contract-construction primitive that produces the evidence state consumed by routing and submission.

A. Algorithm

The MPS optimizer solves a bi-objective weighted set cover:

$$\max_{S \subseteq \mathcal{E}} \alpha \cdot \text{cov}_{\mathcal{O}}(S) + (1-\alpha) \cdot \text{cov}_{\mathcal{C}}(S) \quad \text{s.t.} |S| \leq K, \text{tokens}(S) \leq B. \tag{6}$$

At each step, the greedy algorithm selects the evidence node with the highest marginal utility:

$$e^* = \arg\max_{e \in \mathcal{E} \setminus S} \frac{\Delta \text{cov}_{\mathcal{O}}(e|S) + \lambda \cdot \Delta \text{cov}_{\mathcal{C}}(e|S)}{\text{tokens}(e)}. \tag{7}$$

B. Theoretical Guarantee

Theorem 1 (Approximation Ratio)**.** *Let $n = \max(P, M)$ where P is the number of obligations and M is the number of claims. The greedy algorithm (Algorithm 1) achieves an approximation ratio of $O(\ln n)$ for the bi-objective weighted set cover problem.*

Proof sketch. The bi-objective function $f(S) = \alpha \cdot \text{cov}_{\mathcal{O}}(S) + (1-\alpha) \cdot \text{cov}_{\mathcal{C}}(S)$ is monotone submodular, as both coverage functions are submodular and a non-negative weighted sum of submodular functions is submodular. By the classical result of Nemhauser, Wolsey, and Fisher [10], the greedy algorithm for monotone submodular maximization under cardinality constraint achieves a $(1 - 1/e)$-approximation. For the weighted set cover formulation, the approximation ratio is $H_n = \sum_{i=1}^{n} 1/i = O(\ln n)$, where H_n is the n-th harmonic number [11]. □

IV. Deficit-Guided Escalation

After the initial contract is built, the system must decide whether additional verification or graph rescue is worth its cost. CPN supplies the intervention primitive: it measures which evidence paths are necessary and whether a proposed escalation can replace a weak path without losing supported content. In the frozen SAGE implementation, the learned router estimates severe contract gaps, while the deterministic guard enforces preservation after repair.

A. Intervention Framework

For each $e_j \in S^*$, we perform a counterfactual intervention: remove e_j and re-run the MPS optimizer on $\mathcal{E} \setminus \{e_j\}$. Let S_{-j} denote the resulting subgraph. We measure:

- **Claim loss**: $\Delta_C(e_j) = \text{cov}_{\mathcal{C}}(S^*) - \text{cov}_{\mathcal{C}}(S_{-j})$
- **Obligation loss**: $\Delta_O(e_j) = \text{cov}_{\mathcal{O}}(S^*) - \text{cov}_{\mathcal{O}}(S_{-j})$
- **Sufficiency loss**: $\mathbb{1}[\text{proof_sufficient}(S^*) \wedge \neg\text{proof_sufficient}(S_{-j})]$
- **Objective loss**: $\Delta_f(e_j) = f(S^*) - f(S_{-j})$

B. Classification

Based on the loss patterns, each evidence path is classified into one of five categories: *claim-essential* ($\Delta_C > 0$), *sufficiency-critical* (sufficiency loss), *obligation-critical* ($\Delta_O > 0$), *efficiency-critical* ($\Delta_f \geq 0.05$), or *replaceable* (all losses negligible).

Theorem 2 (Necessity Certification Correctness)**.** *Let ϵ be the coverage tolerance. The CPN certification has a false-positive rate (classifying replaceable evidence as necessary) of at most ϵ, and a false-negative rate (classifying necessary evidence as replaceable) of* 0*.*

Proof sketch. The false-negative rate is 0 because the leave-one-out intervention exactly tests the necessity condition from Definition 3. The false-positive rate arises only when the re-optimization finds an alternative evidence path that provides equivalent coverage within tolerance ϵ, which occurs with probability at most ϵ under the coverage gap assumption. □

V. Controlled Repair and Submission

The final layer converts the contract state into an operational action: keep the text answer, escalate retrieval, accept a bounded repair, or abstain. RCS is the decision interface for this contract. The preservation guard is deliberately deterministic and reference-free at runtime; it can reject a repair that loses supported content even when the repair appears fluent.

A. Risk Score Formulation

Following Definition 4, the calibrated risk is:

$$\begin{aligned} R(a, S) = {} & 0.34 \cdot r_{\text{audit}} + 0.26 \cdot r_{\text{unsupported}} + 0.18 \cdot r_{\text{proof_gap}} \\ & + 0.10 \cdot r_{\text{necessity}} + 0.07 \cdot r_{\text{contradiction}}. \end{aligned} \tag{8}$$

The submission policy applies hierarchical thresholds:

- **Submit**: $R \leq 0.22$, $\text{cov}_{\mathcal{C}} \geq 0.95$, $\text{cov}_{\mathcal{O}} \geq 0.66$
- **Submit with monitoring**: $R \leq 0.42$, $\text{cov}_{\mathcal{C}} \geq 0.80$
- **Block**: unsupported claims > 0 or contradicted claims > 0
- **Defer/abstain**: otherwise

B. PAC-Bayesian Generalization Bound

Theorem 3 (Generalization Bound)**.** *Let $\mathcal{D}$ be the distribution over question-evidence pairs, m the sample size, and $\delta \in (0, 1)$. With probability at least $1 - \delta$ over the draw of m i.i.d. samples, for any submission policy π in the policy class Π:*

$$\mathbb{E}_{(q,\mathcal{E})\sim\mathcal{D}}[\textit{Untrust}(\pi(q, \mathcal{E}))] \leq \hat{R}_{\textit{emp}}(\pi) + \sqrt{\frac{\text{KL}(\pi \| \pi_0) + \ln(2m/\delta)}{m}}, \tag{9}$$

where $\hat{R}_{\textit{emp}}(\pi)$ is the empirical risk, $\text{KL}(\pi \| \pi_0)$ is the KL divergence from the prior policy π_0, and Untrust$(\cdot)$ measures answer untrustworthiness.

Proof sketch. This follows from the McAllester PAC-Bayesian theorem [12], applied to the policy class Π of threshold-based submission rules parameterized by (τ_R, τ_C, τ_O). The calibrated risk score $R(a, S)$ provides a computable proxy for $\hat{R}_{\text{emp}}(\pi)$. The policy class complexity is bounded by $\text{KL}(\pi \| \pi_0) \leq O(d \ln(m/d))$ where $d = 3$ is the number of threshold parameters. □

VI. Experiments

A. Datasets and Protocol

The current experimental package separates mechanism validation from frozen holdout evaluation. UltraDomain Mix130 is used for strict VSR ablation and candidate replay. Legal, Biology, and Domain-windowed slices provide external retrieval-baseline comparisons. Locked unseen75 and strict Repeat1–8 holdouts are source-line/source-context disjoint from prior development surfaces and are used only after freezing SAGE v33-preservation.

TABLE I
Current Evaluation Surfaces

Surface	Questions	Role
UltraDomain Mix130	130	Strict VSR ablation and candidate replay
Legal/Biology/Domain external slices	120	Retrieval-baseline and semantic judge comparison
Locked unseen75	75	Primary post-freeze zero-overlap holdout
Locked Repeat1–8	251	Supplementary zero-overlap robustness and stress holdouts

B. Baselines

We compare against **Naive RAG** and **HyDE** under same-context, same-metric settings. We also report **GraphRAG-lite** as a lightweight community graph baseline and **LightRAG current-130** as the evaluated representative heavy graph-RAG comparator. SAGE v33-preservation is treated as the frozen lead instantiation rather than a post-hoc tuned variant.

C. Metrics

We report lexical F1 and ROUGE-L, faithfulness, same-rubric LLM judge scores, pairwise A/B target win rates across five qualitative dimensions, runtime per question, and coverage/completion status. We keep lexical overlap and semantic judge results separate because they reward different answer properties.

D. Results

Tables II–IX are generated from completed artifacts by `experiments/build_paper_latex_tables.py`; the corresponding Markdown/JSON package is stored under `results/paper_tables/`.

E. Interpretation

The strict ablation supports the VSR mechanism claim: verification, repair, and rescue improve the internal no-repair baseline and materially reduce abstention on Mix130. The stronger paper-facing method is SAGE v33-preservation, which is reproducible, frozen before locked holdout evaluation, and safer than the exploratory v56 branch despite v56's aggregate replay gains.

Across locked unseen75 plus Repeat1–8, SAGE v33 improves weighted lexical F1 modestly and improves semantic judge quality substantially. Repeat8 is intentionally reported as a stress slice: it exhausts the remaining strict legal capacity, ties HyDE lexically, and trails Naive lexically, but it does not change the complete 326-row meta conclusion. We therefore frame the main claim as semantic-quality and preference improvement under strict holdouts, not as blanket lexical superiority over every retrieval baseline.

F. Mechanism Depth Audit

To test the central evidence-contract hypothesis directly, we aligned the frozen SAGE v33 Mix130 rows with the same-row 'full_v2_rescue' outputs and computed a reference-free contract deficit from answer-shape and term-coverage signals. This is an observational audit rather than a randomized treatment comparison.

The audit gives a useful negative result. Route-selected rows have larger estimated contract gaps and higher router probabilities, but they are also harder rows: their mean SAGE–VSR F1 delta is lower than for route-skipped rows. Router probability correlates negatively with the observed SAGE–VSR F1 delta ($r = -0.3782$), while the

TABLE II
STRICT VSR ABLATION ON ULTRADOMAIN MIX130

Config	Rows	F1	ROUGE-L	Faith.	Repair	Abstain	s/q
w/o_all	130/130	0.2276	0.1658	0.5000	0.0%	21.5%	9.0
w/o_evidence_extraction	130/130	0.2285	0.1612	0.5353	66.2%	40.0%	37.2
w/o_verification	130/130	0.2138	0.1552	0.5000	0.0%	18.5%	17.6
w/o_repair	130/130	0.2206	0.1591	0.8516	0.0%	23.1%	36.2
full	130/130	0.2437	0.1749	0.8516	34.6%	27.7%	44.2
full_v2	130/130	0.2824	0.2107	0.9728	40.0%	8.5%	41.0
full_v2_rescue	130/130	0.2878	0.2194	0.9756	27.7%	0.8%	41.7

TABLE III
PAIRED TESTS FOR STRICT VSR ABLATIONS

Comparison	N	F1 Delta	F1 p	Faith. Delta	Faith. p
full - w/o_all	130	+0.0161	0.0277	+0.3516	¡1e-15
full - w/o_evidence_extraction	130	+0.0152	0.0641	+0.3163	1.78e-15
full - w/o_verification	130	+0.0299	5.51e-07	+0.3516	¡1e-15
full - w/o_repair	130	+0.0231	5.18e-05	+0.0000	1.0000
full - full_v2	130	-0.0386	2.51e-05	-0.1211	2.03e-07
full - full_v2_rescue	130	-0.0441	3.46e-05	-0.1239	4.53e-07

simple contract-gap proxy has a weaker negative correlation ($r = -0.1511$). Thus, the current contract mechanism is effective as a difficulty detector, but not yet as an upgrade-utility estimator. This separates two research problems that are often conflated: predicting that an answer is risky versus predicting that paying for escalation will improve it.

The next method iteration should therefore learn or calibrate an impact-aware escalation score, using only reference-free runtime features and validating it on a new locked holdout. The current results do not justify claiming that every high-deficit row benefits from escalation.

VII. RELATED WORK

A. Graph-based RAG

GraphRAG [1] constructs community summaries from entity-relationship graphs for global retrieval. LightRAG [2] employs dual-level retrieval combining entity and relationship matching. HippoRAG [3] uses personalized PageRank for multi-hop reasoning. None of these approaches formally verify whether the retrieved evidence constitutes a sufficient proof.

B. Evidence Verification in RAG

Self-RAG [4] introduces self-reflection tokens for retrieval gating. RAGAS [5] provides automated evaluation metrics. ARES [6] offers automated RAG evaluation. These approaches evaluate answer quality post-hoc but do not construct formal proof structures.

C. Proof-based Verification

EntailmentBank [7] generates multi-hop entailment trees. FOLIO [8] provides first-order logic reasoning. ProofWriter [9] performs natural language theorem proving. VSR-PathRAG is the first to combine formal proof structure with counterfactual necessity analysis and risk-calibrated submission gating.

VIII. CONCLUSION

We presented VSR-PathRAG as an evidence-contract-guided adaptive RAG framework. MPS, CPN, and RCS are unified as contract construction, deficit-guided escalation, and controlled submission primitives. The completed experiment package shows that the rescue-enhanced VSR layer improves the internal no-repair baseline, while the frozen SAGE v33-preservation instantiation improves semantic judge quality and pairwise preference on 326 strict

TABLE IV
EXTERNAL RETRIEVAL BASELINES AND SAGE V33

Dataset	N	Naive	HyDE	VSR Rescue	SAGE v33	Best	SAGE-Best
Legal full context	20	0.3632	0.3760	0.3496	0.3925	hyde	+0.0164
Biology full context	50	0.2365	0.2385	0.2287	0.2385	hyde	+0.0000
Domain expansion windowed	50	0.2711	0.2549	0.2337	0.2881	naive	+0.0170

TABLE V
LOCKED HOLDOUT WEIGHTED LEXICAL RESULTS, UNSEEN75 PLUS REPEAT1-8

Method	N	F1	ROUGE-L	s/q	Chars
naive	326	0.2972	0.2338	14.2	665.9
hyde	326	0.2972	0.2317	25.1	687.6
sage_v33_preservation_delta	326	0.3164	0.2532	25.1	764.3

locked-holdout rows. The mechanism audit also exposes the next scientific bottleneck: current contract-gap routing detects difficult rows, but its score is negatively associated with realized upgrade benefit. Future work should therefore target impact-aware escalation, official large-scale graph-baseline reruns, broader standard-benchmark validation, and human evaluation.

LIMITATIONS

1) **LLM dependency**: Claim extraction and evidence signing rely on LLM quality.
2) **Scalability**: CPN's $O(|S| \times \text{MPS})$ complexity limits large $|S|$.
3) **Domain generalization**: The current strongest locked evidence comes from UltraDomain-derived domain slices; broader standard-benchmark validation remains future work.
4) **Baseline comparability**: GraphRAG-lite and LightRAG current-130 provide completed representative graph-RAG evidence, but full official graph-baseline reruns remain approximate or budget-dependent.
5) **Metric tension**: SAGE v33 is strong on same-rubric judge and pairwise preference, while lexical F1 gains over Naive RAG and HyDE are modest and not uniformly positive on every stress slice.
6) **Risk–benefit mismatch**: The current reference-free contract-gap router identifies difficult rows, but Mix130 analysis shows that high routing probability does not imply larger realized improvement after escalation.
7) **Theory**: The PAC-Bayesian bound assumes i.i.d. sampling, which may not hold for conversational RAG.

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TABLE VI
LOCKED HOLDOUT WEIGHTED JUDGE RESULTS, COMPLETE 326-ROW SUBSET

Method	N	Correct	Coverage	Overall
naive	326	3.092	2.856	2.960
hyde	326	3.110	2.856	2.942
sage_v33_preservation_delta	326	3.816	3.469	3.555

TABLE VII
LOCKED HOLDOUT WEIGHTED PAIRWISE WIN RATES, COMPLETE 326-ROW SUBSET

Comparison	N	SAGE Target Win Rate
sage_v33_vs_hyde	326	59.66%
sage_v33_vs_naive	326	61.50%

TABLE VIII
REPEAT8 LEGAL-ONLY EXHAUSTIVE STRESS HOLDOUT

Method	N	F1	ROUGE-L	Judge	s/q
naive	54	0.2958	0.2344	2.444	13.4
hyde	54	0.2862	0.2174	2.352	29.4
sage_v33_preservation_delta	54	0.2862	0.2174	2.407	29.4

Repeat8 exhausts the remaining strict legal holdout capacity under the max-two-samples-per-context policy.

TABLE IX
COST AND LATENCY SUMMARY

Dataset	N	Naive s/q	HyDE s/q	VSR s/q	VSR/Naive	VSR/HyDE	VSR-Best F1
Legal full context	20	6.0	14.4	59.6	9.89	4.14	-0.0264
Biology full context	50	6.8	20.7	46.9	6.88	2.26	-0.0098
Domain expansion windowed	50	5.6	14.7	43.3	7.66	2.94	-0.0374

TABLE X
EVIDENCE-CONTRACT MECHANISM AUDIT ON MIX130

Group	N	Mean F1 Delta	Wins/Losses	Mean Router Probability
All aligned rows	130	+0.1768	113/17	0.2440
Route skipped	87	+0.2089	83/4	0.0956
Route selected	43	+0.1119	30/13	0.2991